



**US Army Corps
of Engineers®**
Seattle District



HIRAM M. CHITTENDEN LOCKS

SMALL LOCK

EMERGENCY CLOSURE SYSTEM

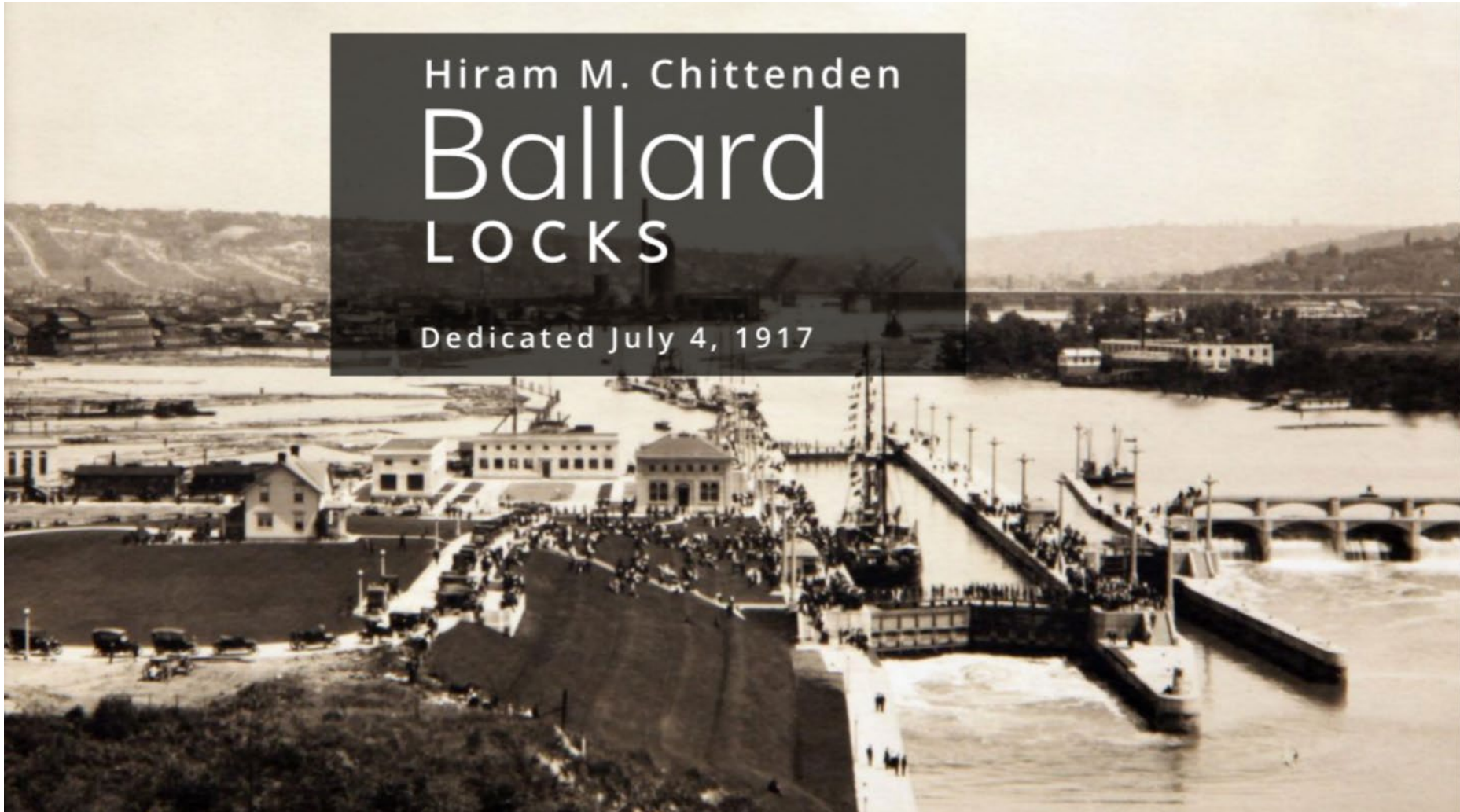
JUNE 17, 2025



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HIRAM M. CHITTENDEN LOCKS (BALLARD LOCKS)



HIRAM M. CHITTENDEN LOCKS

Two locks:

Small Lock 30 x 150 ft

Large Lock 80 x 825 ft

235-foot Dam Spillway

Fish ladder



HIRAM M. CHITTENDEN LOCKS

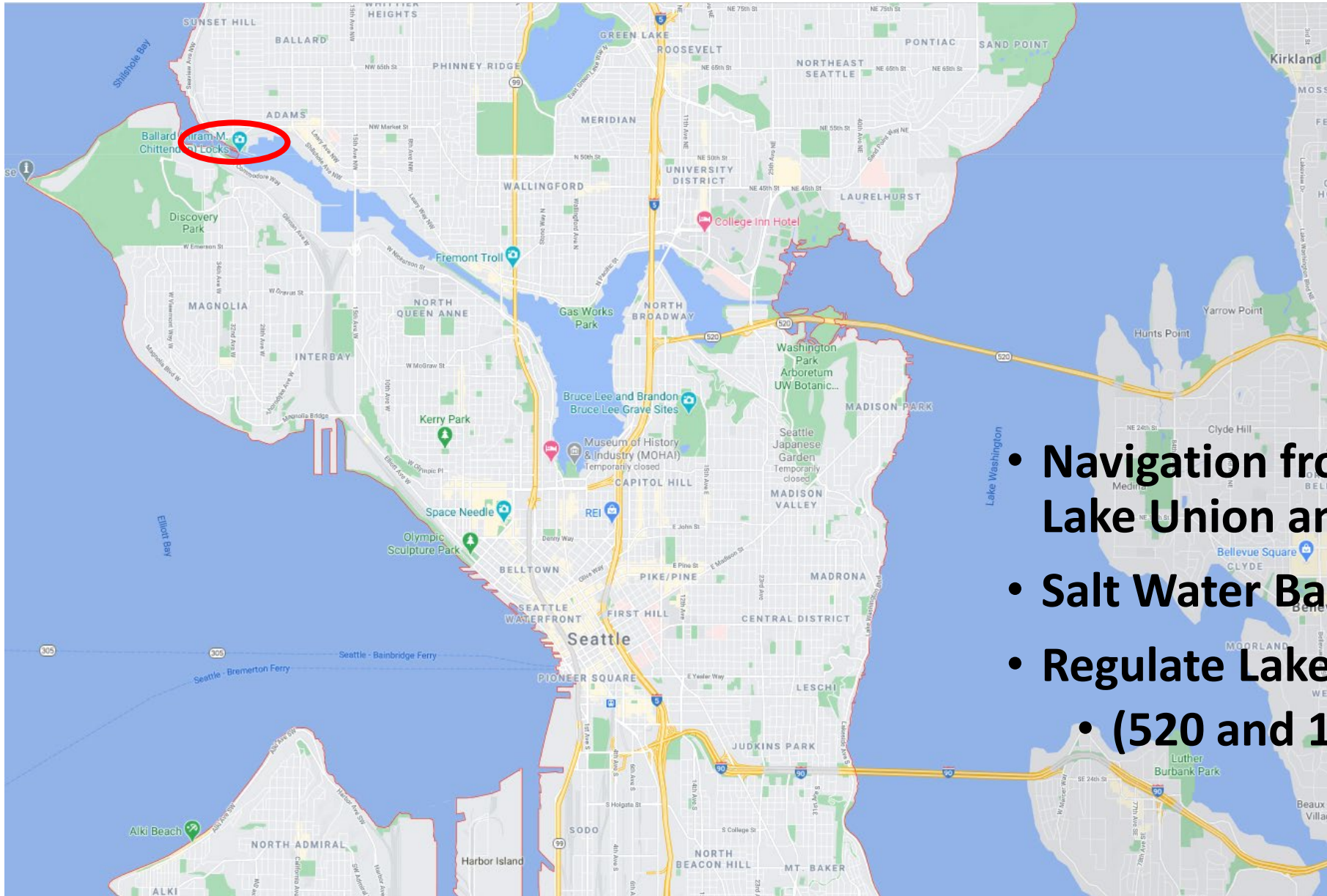
SMALL LOCK



- Cycles up to 50 times per day (summer)
- About 10,000 lockages per year
- More than any lock chamber in the nation



Lock Project Serves Several Purposes



- Navigation from Puget Sound to Lake Union and Lake Washington
- Salt Water Barrier
- Regulate Lake Levels
- (520 and 1-90 Floating Bridges)

Purpose - Regulate Lake Levels



TETRA TECH

- Lake Washington Floating Bridges
- Longest Floating Bridges in the World
 - 520 Constructed in 2016 (\$4.5B)
 - Adding light rail to I-90



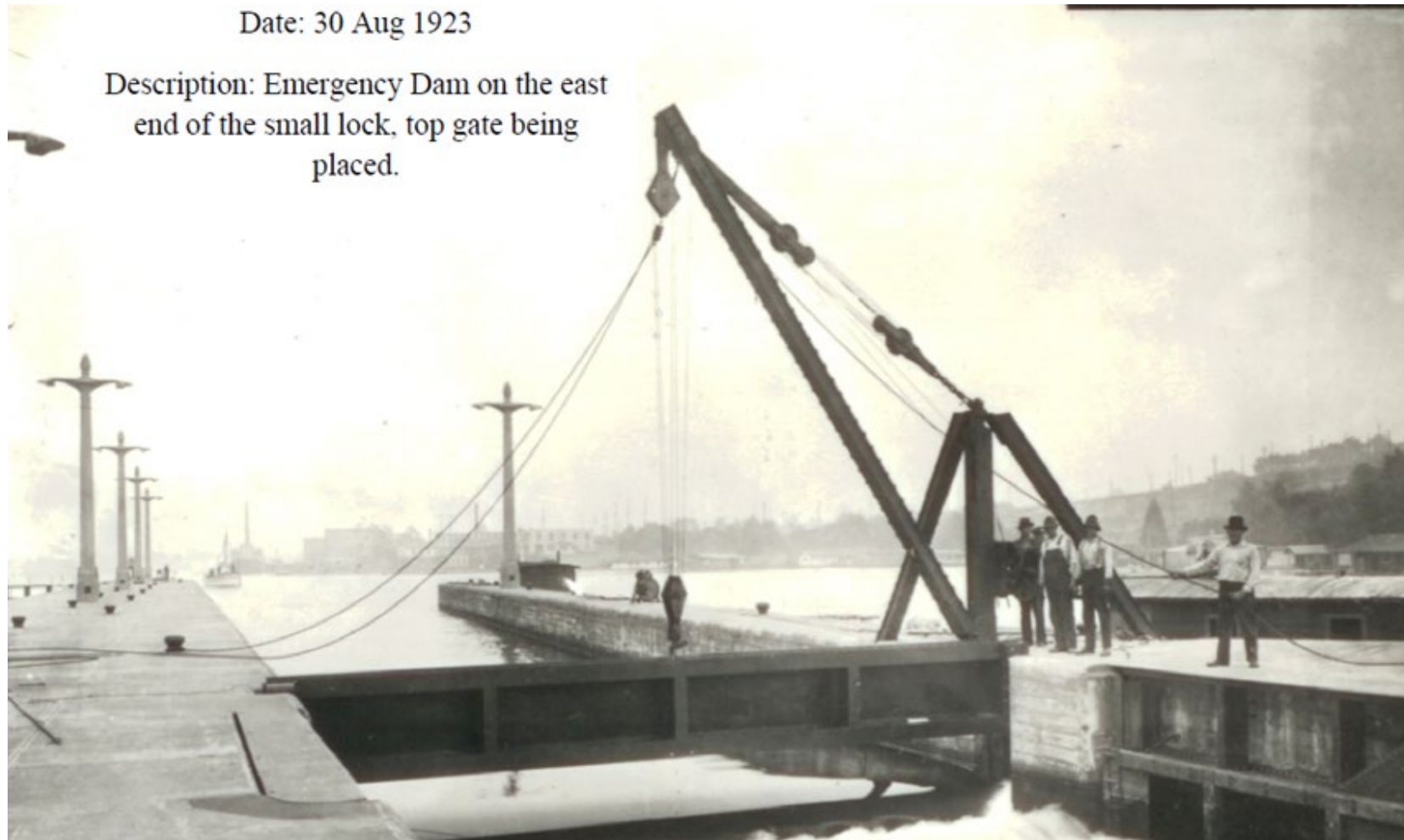
I-90 Floating Bridge

520 Floating Bridge



Existing Emergency Closure System

Constructed in 1917 as part of Original Lock Construction
Has never been used



Existing Emergency Closure System

An emergency closure structure is any device designed to be placed in rapidly flowing water under differential head to prevent the loss of pool if the lock miter gates are damaged and can no longer function to control the pool.

- 5 Stoplogs
- Requires Divers to Install
- Cannot be Placed In Flowing Water
- Seals Leak
- Cannot be used as a work gate
- Crane Undersized

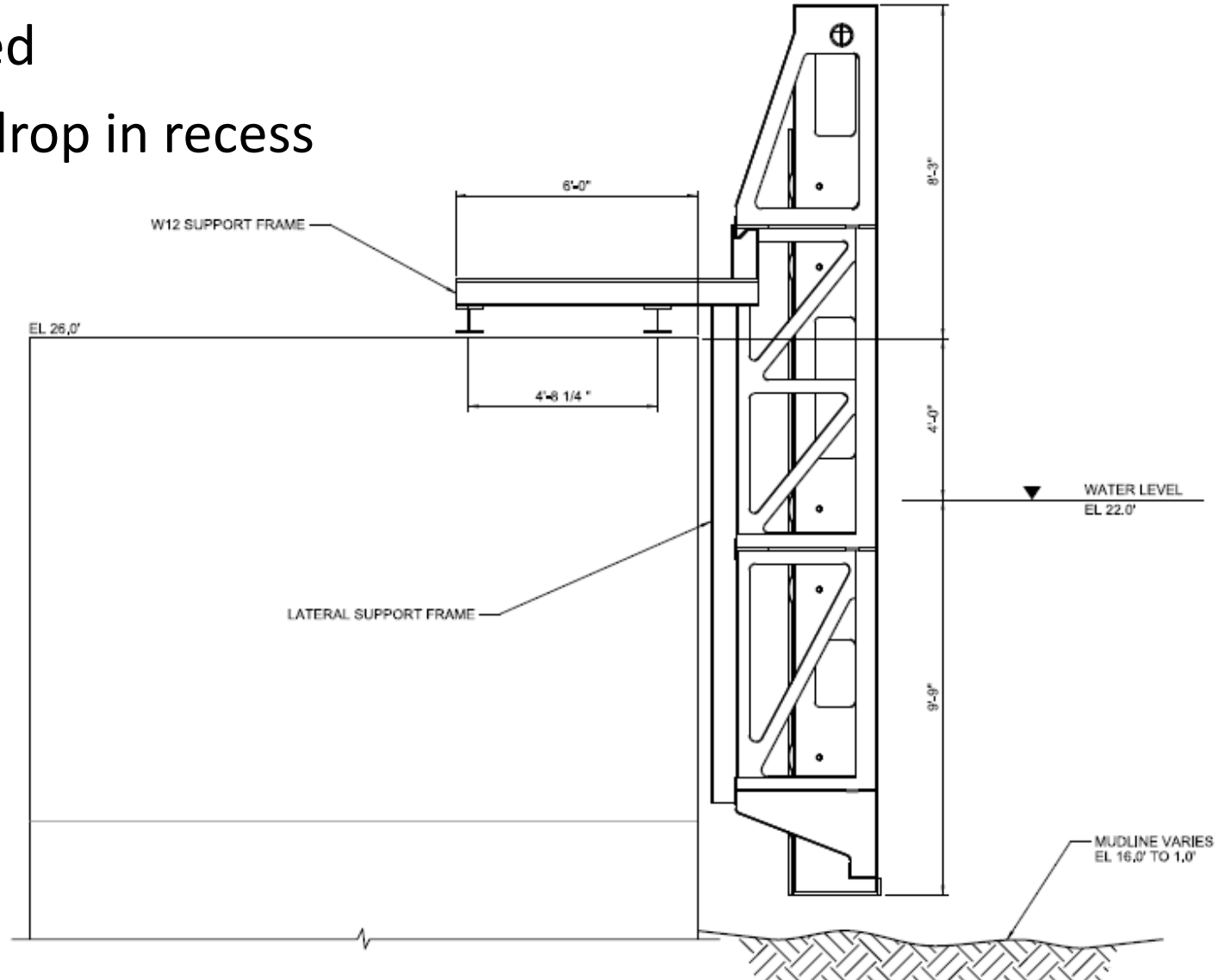
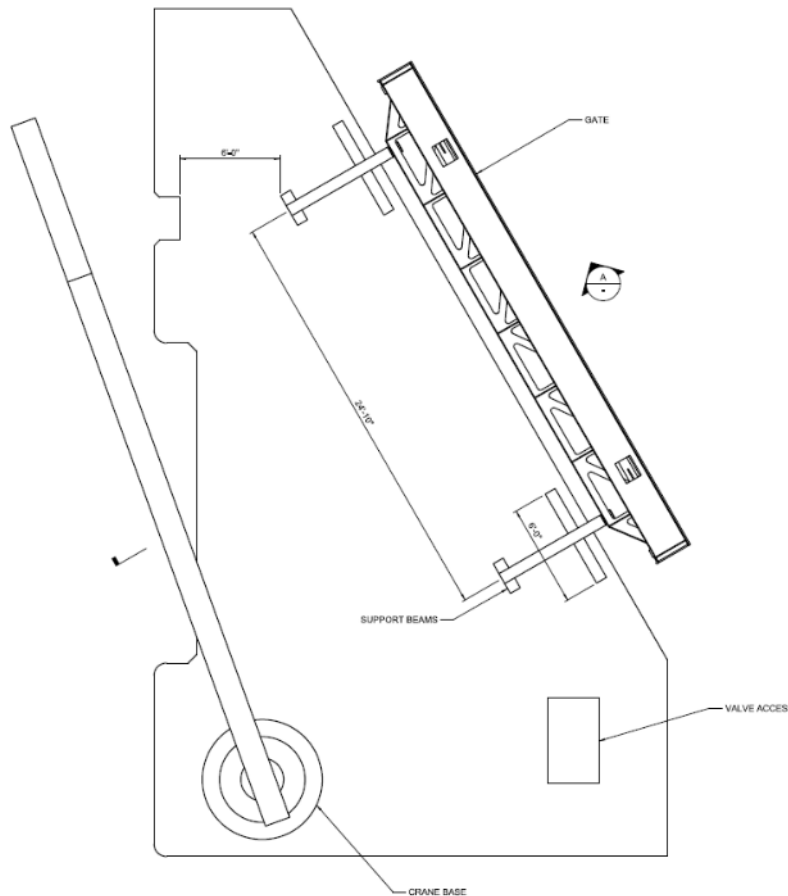


Vertical Storage



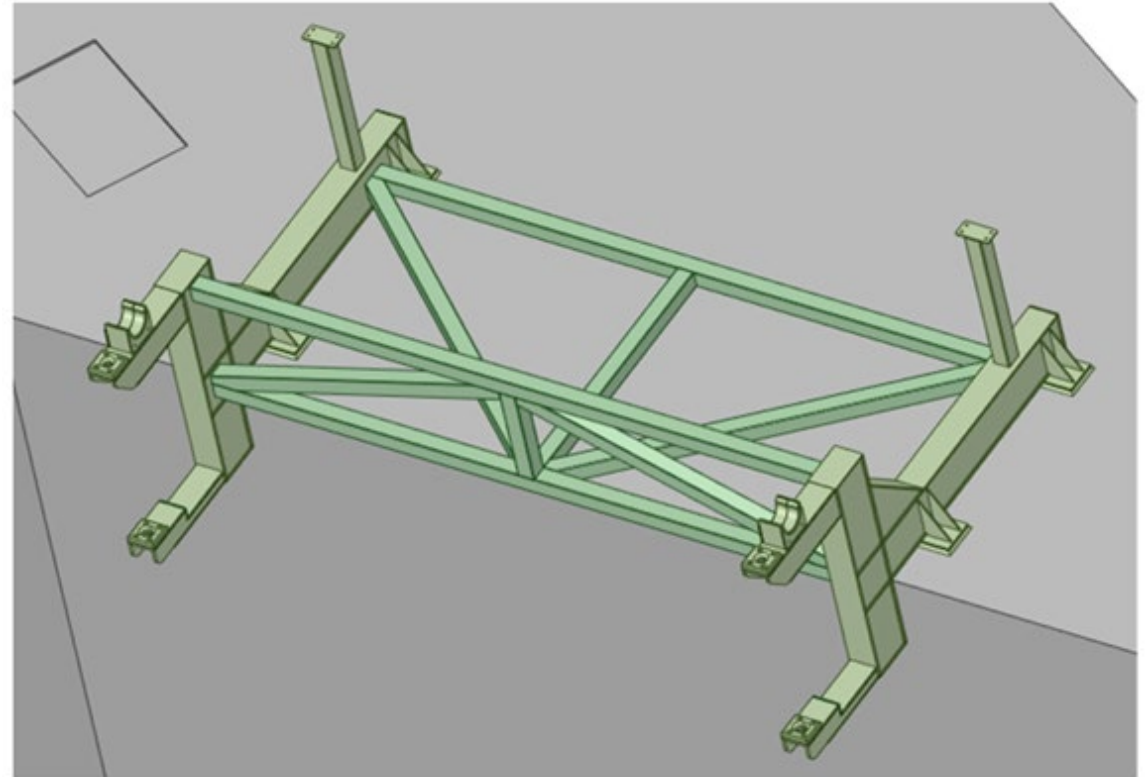
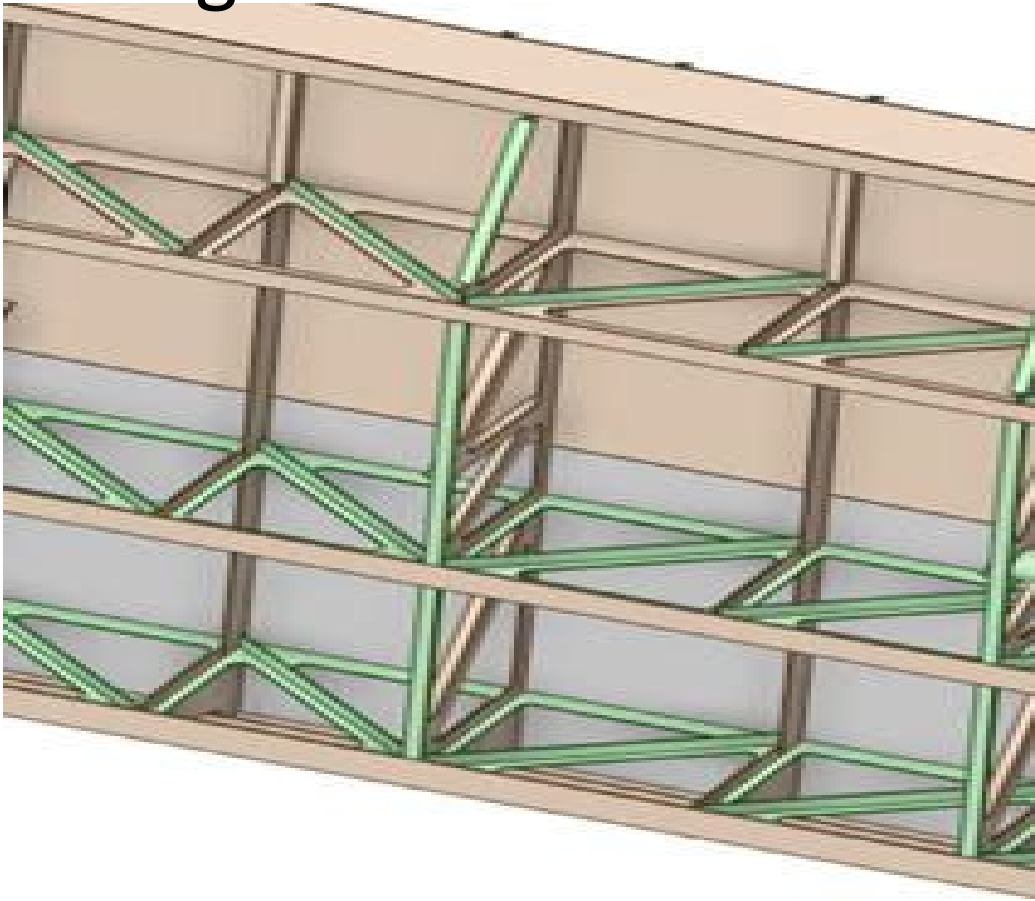
TETRA TECH

- Storage frame mounted to top of monolith (no in-water construction)
- About 10 feet of gate is submerged
- Easy to install - lift vertically and drop in recess



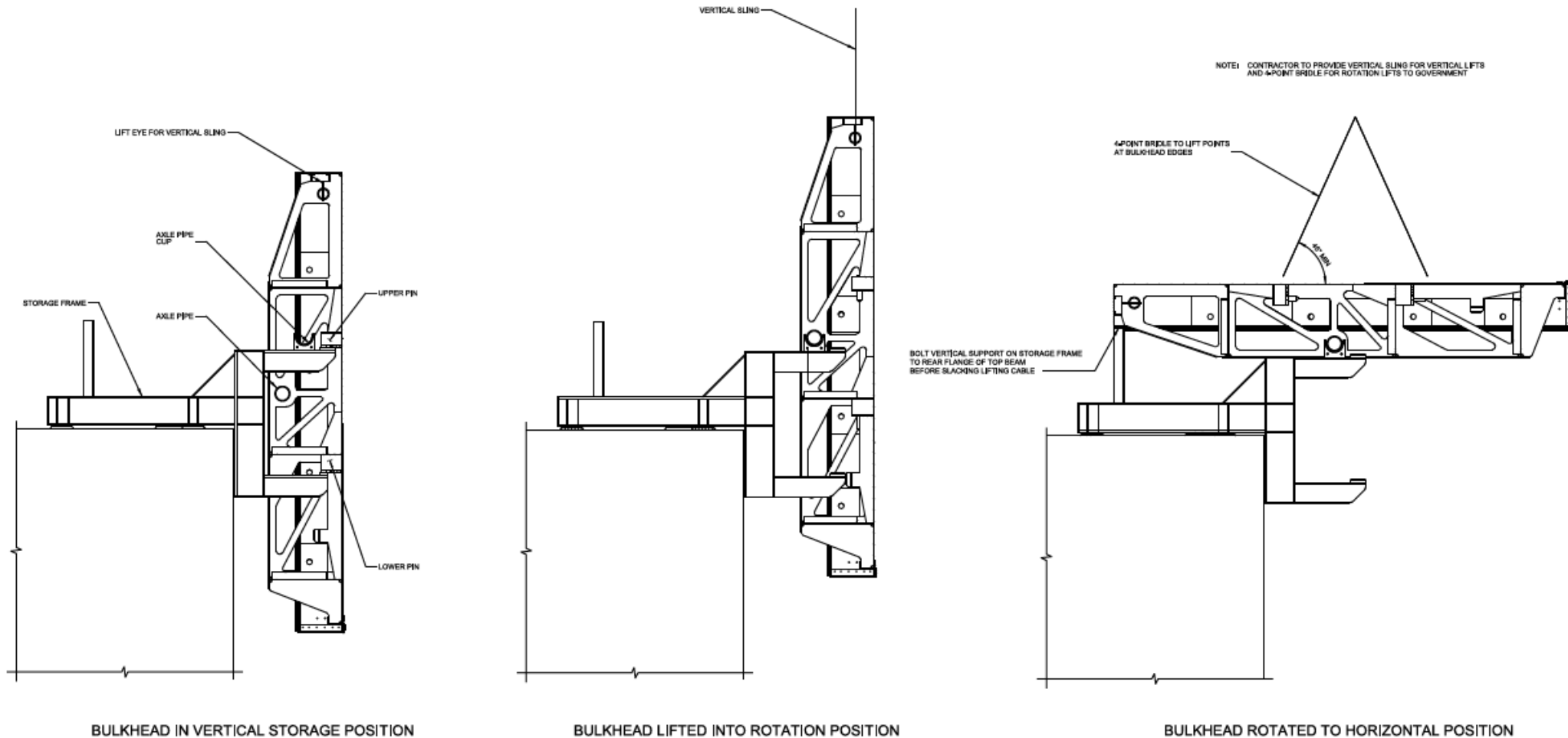
Key Features

- Storage and Ease of Use Critical Factors
- Full height bulkhead with vertical storage is easier to use as a working gate and as an emergency bulkhead
- Larger crane can be beneficial for other maintenance activities



Key Features Continued...

- Provided bulkhead pipe support and storage frame saddles.
- Allows bulkhead to rotate horizontal for minor routine inspection/maintenance.
- Corrosion Protection System with magnesium anodes designed by Norton Corrosion.

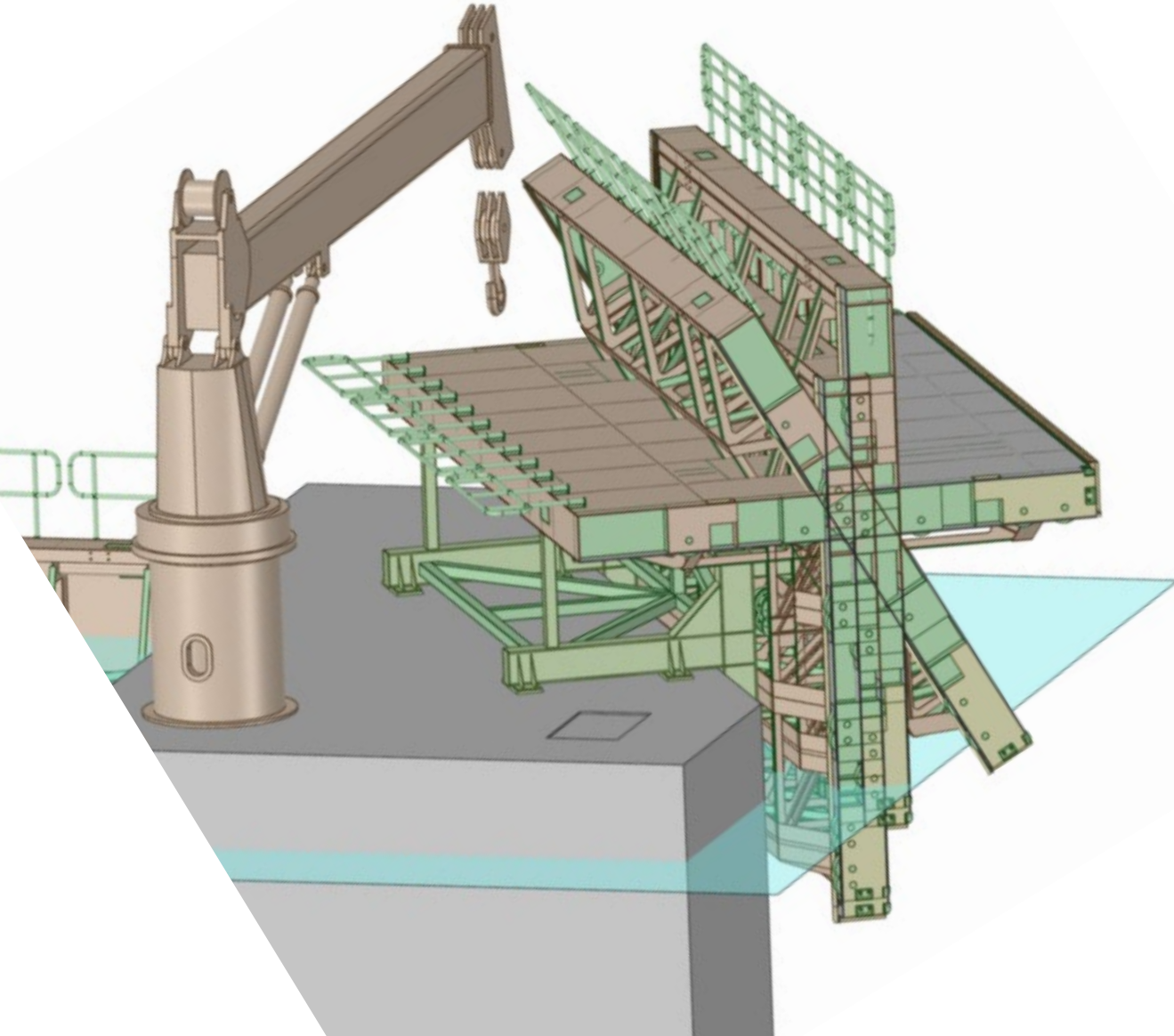


Design with 3D Modeling

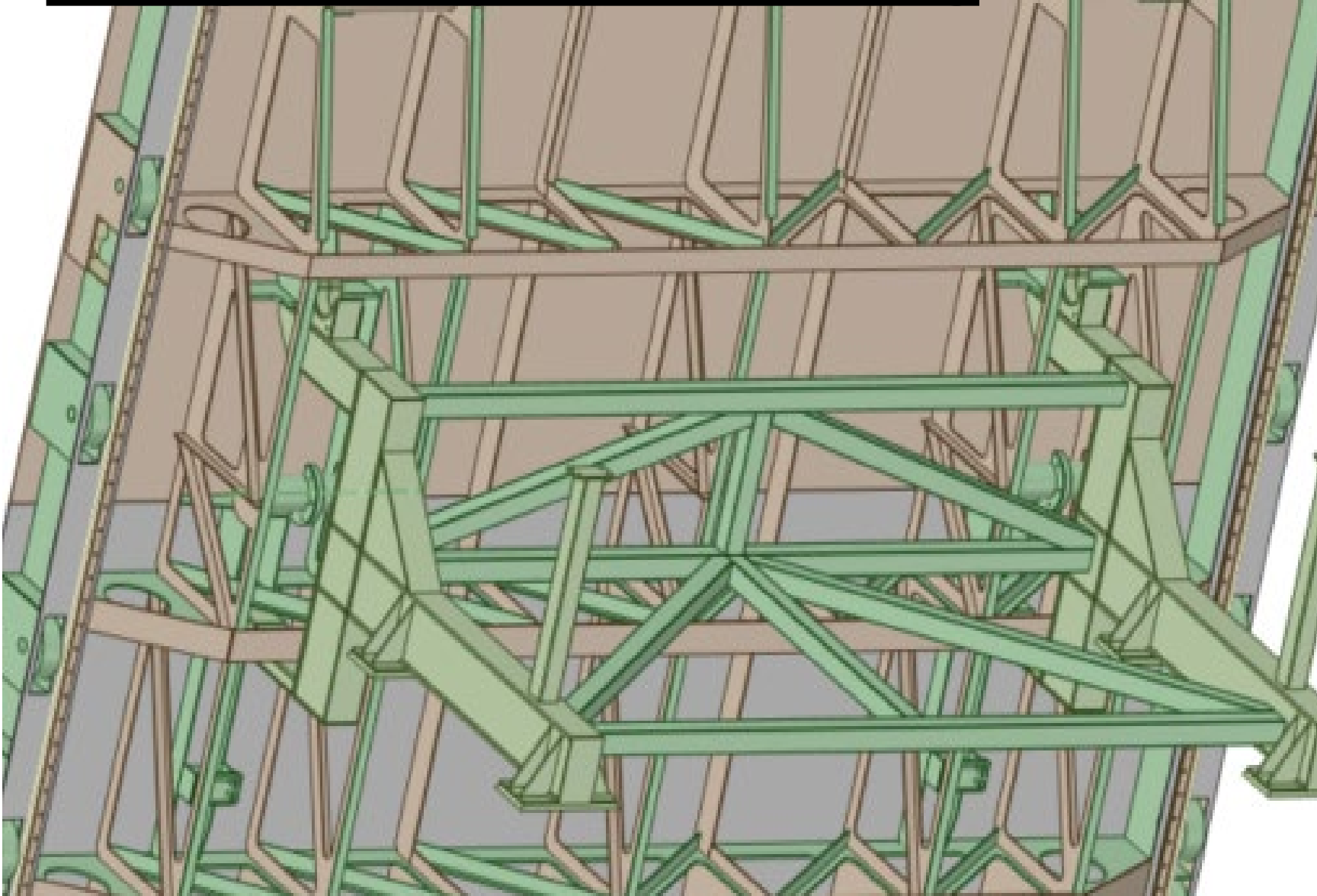
3D Modeling

- Complicated Geometry
- Multiple Positions
- Rotation
- Maintenance Position

Much Easier with 3D
Fewer Errors

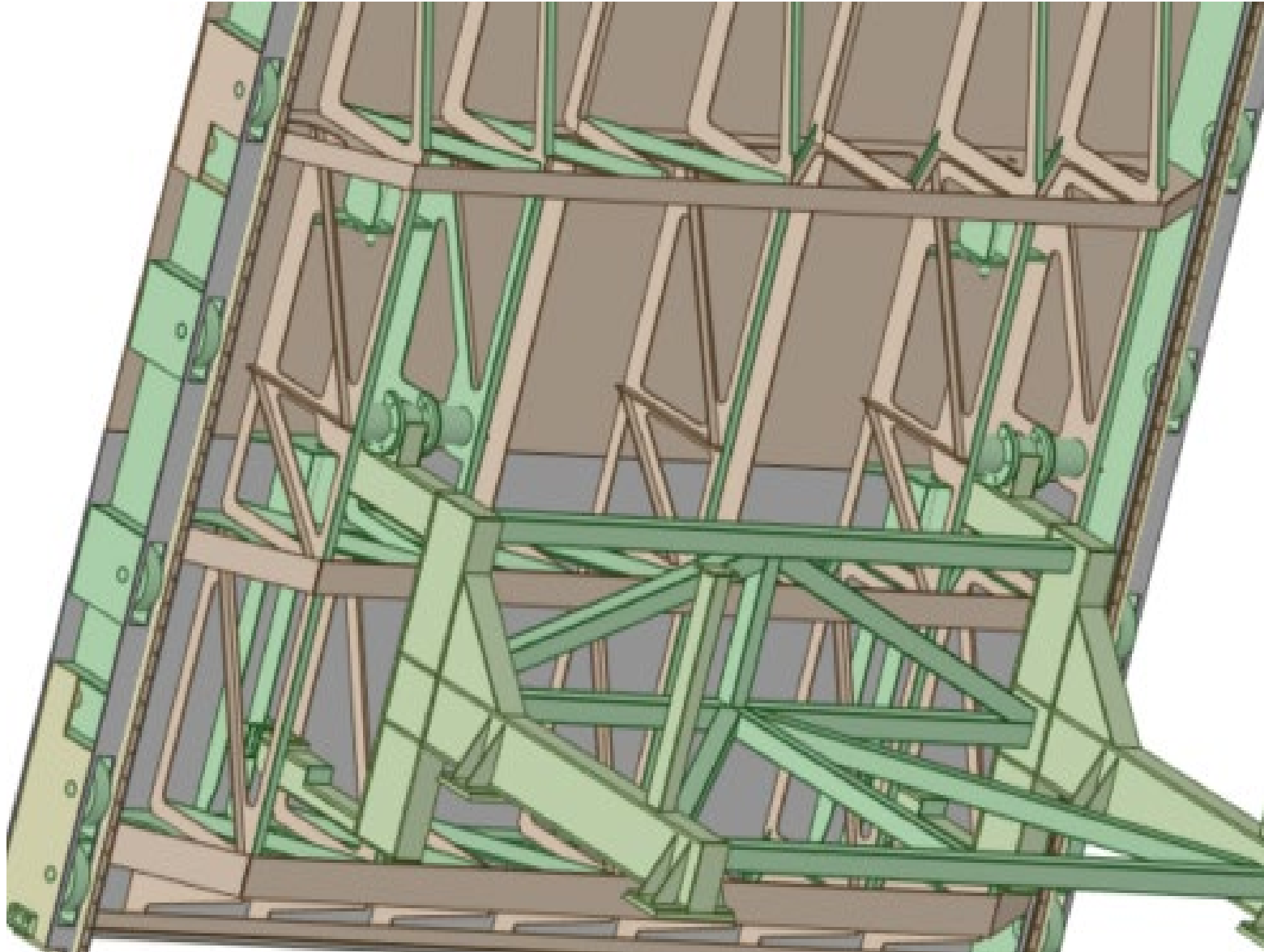


Design with 3D Modeling



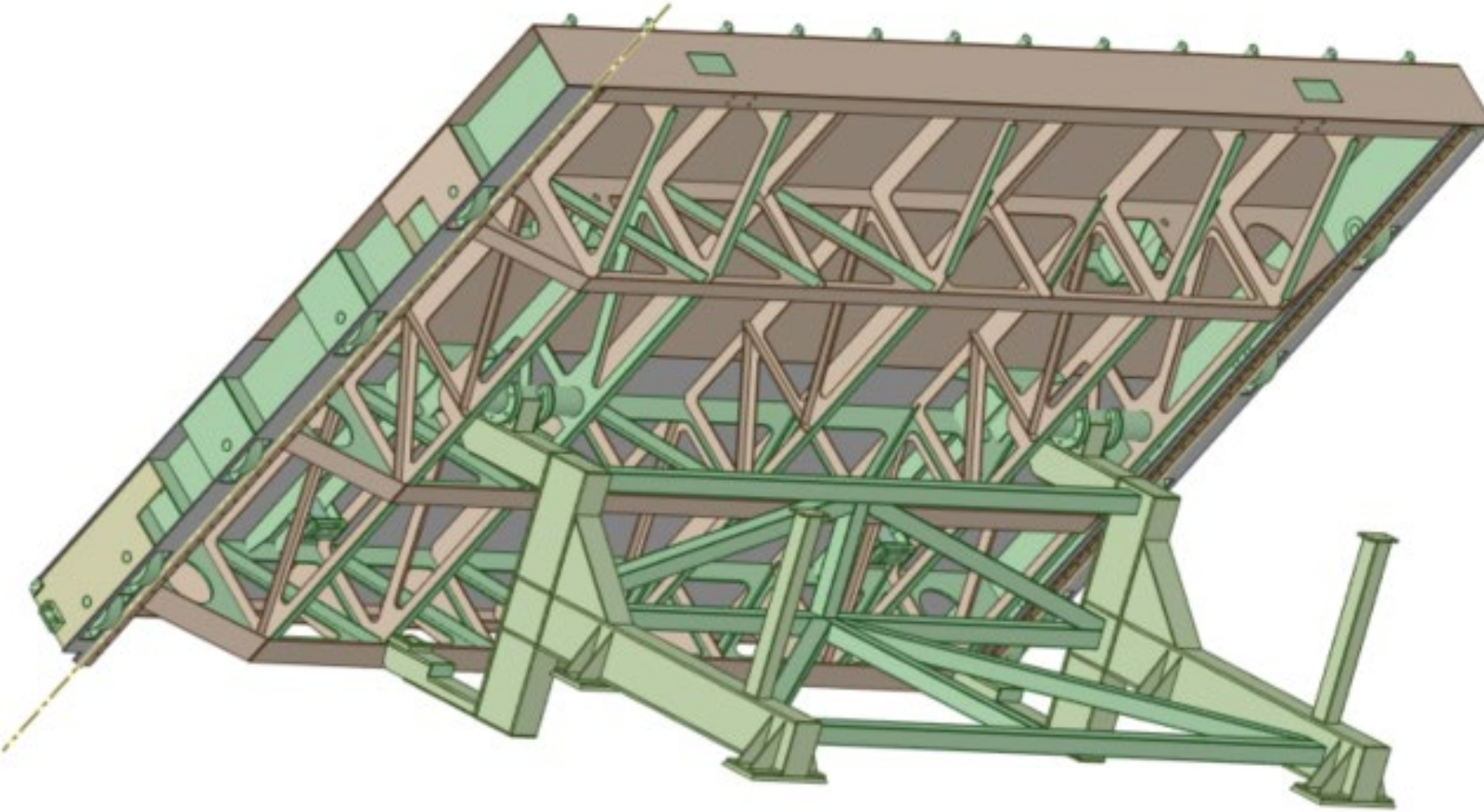
Bulkhead fits on support frame at multiple locations and allows rotation on support pins.

Design with 3D Modeling



Bulkhead
supported on
trunions

Design with 3D Modeling



Bulkhead
being
rotated

Design with 3D Modeling

Bulkhead in
maintenance
position

